Project Plan

Phase 1: Modeling and Simulation

- Assumption: Keep Model with enough fidelity for accurate simulation and simplicity
    - Rigid, linear viscous damping, 1 Input – 2 Outputs, published parameters are true, linearize model.
    - Discrete sensor with realistic resolution, Motor saturation.
- Approach: Study and refer to supplementary material to derive system dynamics
    - Study Ch. 2-5 in Feedback Systems, paper about system dynamics.
    - If Simulink simulation isn't like Qube Servo's. Double check parameters/increase fidelity. Allow for intractability/parametric.

Phase 2: Balancing and Swing-Up

- Assumption: System is fully controllable and observable.
    - Use state observers to supply full state feedback. Use switching to change controllers. System is stable even with noise and disturbances.
- Approach: Study, then create two controllers, prove characteristics of system.
    - Study Ch. 6-7 & 12 in Feedback Systems, paper about swing up controllers
    - Compare true simulated results with estimated velocity from observer
    - Troubleshoot state machine using simulation results
    - Add potential disturbances, sensor noise, actuator constraints, then analyze

Phase 3: Robustness Analysis

- Assumption: Parametric uncertainty will exist. High Robustness is possible.
- Approach: Apply Variation, analyze behavior, prepare sim-to-real transfer
    - Study Ch. 9,11,12 in Feedback System
    - Higher fidelity: parametric variation, static friction, & joint damping,
    - Stability Margins, Monte Carlo, sensitivity analysis for sim-to-real transfer.

Phase 4: Learning-based Control

- Assumption: Simple learning-based controller will control the system. LQR is Baseline
- Approach:
    - Study Ch. 1-4 in RL and Optimal Control
    - Try multiple ML Models and compare with LQR baseline.
    - Finally compare with Qube Servo RL Example in MATLAB.